

Binomial Theorem: An extending intuition to combinatorics

Focus: Binomial Expansions and Set-Theoretic Polynomial Combinatorics

1. Decompressing the Notation: Frequency vs. Valuation

When preparing for engineering or quantitative analysis (including the HKDSE Module 1/2 spectrum), think of mathematical structures as computational processes rather than static formulas.

Core Concept: The Set-Theoretic View of ax^b

When we write a term ax^b , we are looking at a shorthand code for an element and its count within a database:

- **Addition (+):** Acts as a catalog aggregator (counting distinct options or varieties).
- **Multiplication ($\times$):** Acts as a collection builder (drawing combinations together).

Hence, a linear expression $ax+b$ is a **compressed set representation** storing exactly a copies of target variable x , alongside b copies of a baseline identity marker 1.

When computing the n -th power of a system, $(ax + by)^n$, you are choosing exactly one item from each of the n independent parent buckets. To build a composite element with a target signature $x^k y^{n-k}$, you choose x from k brackets, and fill the remaining $n - k$ slots with y . The number of ways to do this is given by the combination formula $C(n, k) = \binom{n}{k}$.

2. Advanced Generalization: Index Tracking via Exponents

If we extend this framework to polynomials $(a + bx + cx^2 + \dots)^n$, the exponents act as trackers for structural weights or event values. When we multiply terms, their exponents add together—acting as an exact mapping for calculating cumulative sums across multi-stage processes.

Example 1: Multi-Option Set Expansion (HKDSE M2 Focus)

Determine the coefficient of x^3 within the algebraic expansion of $(1 + 3x - x^2)^4$.

Solution: Using the general distribution expansion framework, we seek paths from 4 brackets that combine to an exponent sum of 3. Let n_1 be the number of times we choose 1, n_2 be the times we choose $3x$, and n_3 be the times we choose $-x^2$. Our system layout requires:

$$n_1 + n_2 + n_3 = 4 \quad \text{and} \quad 1(n_2) + 2(n_3) = 3$$

Analyzing non-negative integer possibilities for (n_1, n_2, n_3) :

1. Case 1: Set $n_3 = 1 \implies n_2 = 1 \implies n_1 = 2$.
Calculated Combination Count: $\frac{4!}{2!1!1!} (1)^2 (3)^1 (-1)^1 = 12 \times 1 \times 3 \times (-1) = -36$.
2. Case 2: Set $n_3 = 0 \implies n_2 = 3 \implies n_1 = 1$.
Calculated Combination Count: $\frac{4!}{1!3!0!} (1)^1 (3)^3 (-1)^0 = 4 \times 1 \times 27 \times 1 = 108$.

Summing these operational paths yields the final structural coefficient:

$$\text{coef}(x^3) = -36 + 108 = 72$$

HKDSE Exam Tip: Avoiding Coefficient Traps

A common error in public examinations is ignoring the negative sign inside the term $(-x^2)_3^n$. Always capture the sign inside the frequency parameters to prevent systemic sign errors during multinomial expansion.

3. Real-World Systems: Modeling Random Processes (Dice Convolution)

We can easily compute complex sample spaces using polynomial configurations. Take a regular, fair six-sided die. We can store all possible roll profiles inside a single choice polynomial:

$$S(x) = x^1 + x^2 + x^3 + x^4 + x^5 + x^6$$

If you roll n independent dice simultaneously, the joint outcome counting system is completely described by the expanded profile of $S(x)^n$.

Example 2: Probability Derivations (Mocking HKASL M&S 1996 Q3)

Suppose we roll 3 independent fair dice. Calculate the exact structural probability that the sum of the face scores equals 10.

Solution: The total configuration count matching a sum of 10 is equal to $\text{coef}(x^{10} \text{ in } (x + x^2 + x^3 + x^4 + x^5 + x^6)^3)$. Let's factor out the lowest variable choice:

$$S(x)^3 = [x(1 + x + x^2 + x^3 + x^4 + x^5)]^3 = x^3 \left(\frac{1 - x^6}{1 - x} \right)^3$$

We need to find the coefficient of x^{10} in:

$$x^3(1 - 3x^6 + 3x^{12} - x^{18})(1 - x)^{-3}$$

This simplifies to searching for the coefficient of x^7 in the remaining expansion product:

$$(1 - 3x^6 + 3x^{12} - x^{18}) \sum_{k=0}^{\infty} \binom{k+2}{2} x^k$$

Let's extract the valid matching tracks:

- Path A: Multiply $1 \times \text{coef}(x^7) \implies \binom{7+2}{2} = \binom{9}{2} = 36$.
- Path B: Multiply $(-3x^6) \times \text{coef}(x^1) \implies -3 \times \binom{1+2}{2} = -3 \times 3 = -9$.

Total valid configuration counts = $36 - 9 = 27$.

Since the total sample grid size is $6^3 = 216$, the system probability is:

$$P(\text{Sum} = 10) = \frac{27}{216} = \frac{1}{8} = 0.125$$

4. Practice Checkpoints

Exercise 4.1: Structural Parameter Finding

Find the coefficient of x^2 in the algebraic expansion of $(x^2 - \frac{2}{x})^7$.

Solution: The general term format is given by: $T_{r+1} = \binom{7}{r}(x^2)^{7-r}(-\frac{2}{x})^r = \binom{7}{r}(-2)^r x^{14-2r-r} = \binom{7}{r}(-2)^r x^{14-3r}$.

We target an exponent valuation of 2: $14 - 3r = 2 \implies 12 = 3r \implies r = 4$.

Substitute $r = 4$ back into the formula to evaluate the coefficient:

$$\text{coef}(x^2) = \binom{7}{4}(-2)^4 = 35 \times 16 = 560$$

Exercise 4.2: Theoretical Application (The Even-Subgroup Identity)

Prove that the sum of the even-valuation combination parameters matches the sum of the odd-valuation parameters: $\binom{n}{0} + \binom{n}{2} + \dots = \binom{n}{1} + \binom{n}{3} + \dots = 2^{n-1}$.

Solution: Consider the base compressed identity expansion for two competing uniform traits: $(x + y)^n = \sum_{r=0}^n \binom{n}{r} x^{n-r} y^r$.

Set $x = 1$ and $y = -1$. This neutralizes the net valuation balance:

$$(1 + (-1))^n = \sum_{r=0}^n \binom{n}{r} (1)^{n-r} (-1)^r \implies 0 = \binom{n}{0} - \binom{n}{1} + \binom{n}{2} - \binom{n}{3} + \dots$$

Separating the opposing positive and negative identity coefficients yields:

$$\binom{n}{0} + \binom{n}{2} + \binom{n}{4} + \dots = \binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots$$

Since the complete sum across all parameters equals 2^n , each identical half must equal $\frac{2^n}{2} = 2^{n-1}$.